Design and Simulation of a 3-DOF SCARA Robotic Manipulator

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Project Summary

This project focuses on the design and simulation of a 3-Degree-of-Freedom (3-DOF) SCARA robotic manipulator capable of performing an autonomous pick-and-place operation. The primary objective is to model the robot's kinematics, plan a complete pick-and-place task, and demonstrate autonomous execution within a simulation environment.

The manipulator is modelled with two revolute joints for planar motion and one prismatic joint for vertical movement. Forward and inverse kinematics are implemented analytically to compute joint configurations for desired end-effector positions. A fully autonomous task sequence is developed, including approach, pick, lift, transfer, place, and return-to-home motions, ensuring smooth and stable trajectories throughout execution.

The simulation is implemented in MATLAB, integrating 3D visualization, task planning, and control logic. Upon completion of the autonomous sequence, optional manual joint control is enabled for demonstration purposes. In addition, Arduino-based servo control is simulated using Tinkercad to validate the feasibility of mapping kinematic outputs to real hardware actuators.

Overall, the project demonstrates autonomous robotic manipulation, a clear separation between task planning and control, and a practical understanding of industrial pick-and-place systems.

Introduction

Robotic manipulators play a crucial role in modern industrial automation by performing repetitive, precise, and high-speed tasks such as pick-and-place operations, assembly, and material handling. Among various robotic configurations, the SCARA (Selective Compliance Assembly Robot Arm) is widely used due to its simple structure, high rigidity in the vertical direction, and efficient planar motion.

This project focuses on the design, kinematic modelling, and autonomous simulation of a 3-DOF SCARA robotic manipulator. The manipulator is simulated using MATLAB, while hardware feasibility is demonstrated through Arduino and Tinkercad simulation. The system performs a fully autonomous pick-and-place task, starting from a home position and returning to it without any manual intervention.

Phase I: Virtual Modelling

1. Robot Configuration

The manipulator consists of three degrees of freedom:

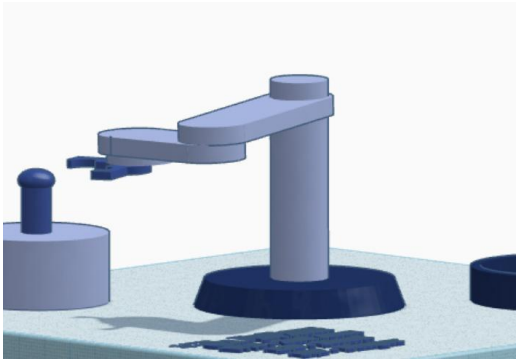
Joint	Type	Function
Joint 1	Revolute	Base rotation (θ_1)
Joint 2	Revolute	Arm extension/rotation (θ_2)
Joint 3	Prismatic	Vertical motion (Z-axis)

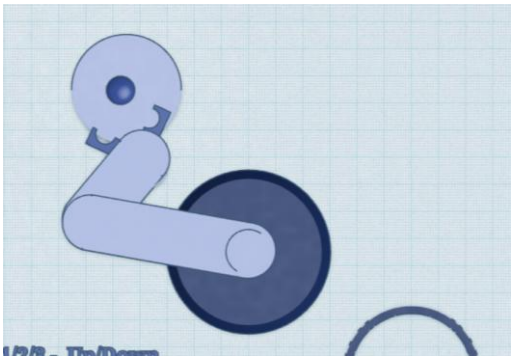
- Two revolute joints for motion in the horizontal (X–Y) plane.
- One prismatic joint for vertical (Z-axis) motion.

This configuration allows efficient positioning of the end effector while maintaining stability during pick-and-place operations.

2. CAD Modelling and Simulation Environment

- The mechanical structure of the SCARA manipulator was modelled using Tinkercad. The design includes the base platform, central vertical column, rotational arm, and end-effector mounting region. The CAD model helps visualize the link lengths, joint placements, and reachable workspace.





- The simulation environment was implemented directly in MATLAB, where the robot geometry, joint motions, and workspace constraints were recreated using 3D visualization techniques. The simulation includes real-time visualization of links and joints, forward and inverse kinematics, autonomous pick-and-place routine, and manual joint control using sliders.

Phase II: Kinematic Solver

1. Forward Kinematics

Forward kinematics was implemented to compute the position of the end effector from given joint variables. Using link lengths $L_2$and $L_3$and joint angles $\theta_1$and θ_2, the end-effector position is calculated as:

$$x = L_2\cos(\theta_1) + L_3\cos(\theta_1 + \theta_2)$$
$$y = L_2\sin(\theta_1) + L_3\sin(\theta_1 + \theta_2)$$
$$z = z$$

These equations were implemented in MATLAB to visualize robot posture and motion.

2. Inverse Kinematics

An analytical inverse kinematics solution was developed to compute the required joint variables for a desired end-effector position Inverse kinematics computes the required joint angles for a desired end-effector position (x, y, z). Workspace feasibility was verified before computing joint values to avoid unreachable configurations.

The kinematic model follows a DH-style revolute–revolute–prismatic joint structure, enabling stable and accurate positioning.

Phase III: Task Planning

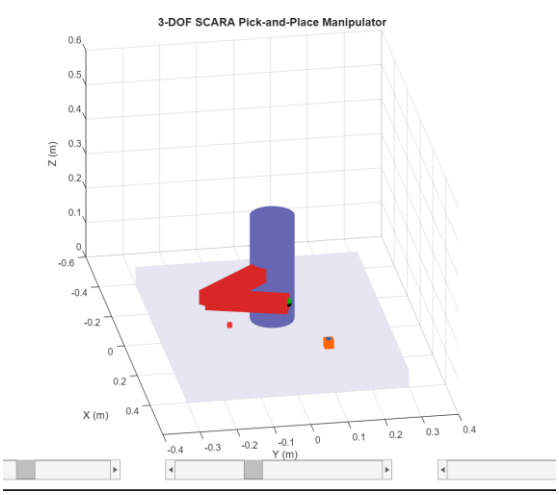
1. Pick-and-Place Task Definition

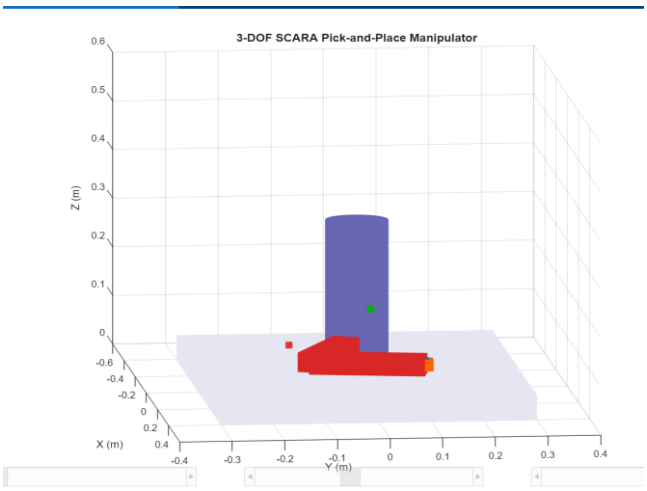
A fully autonomous pick-and-place task was implemented using predefined task points:

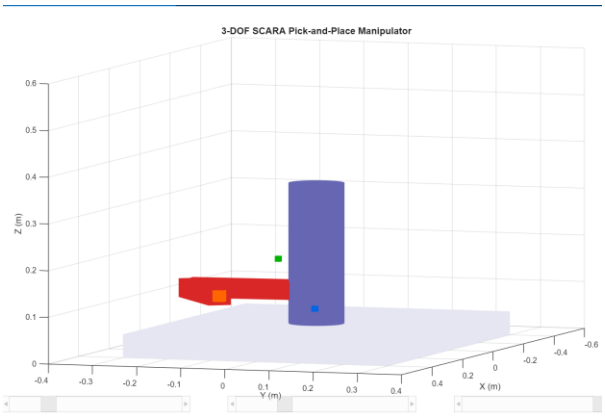
- Home position
- Pick position
- Place position

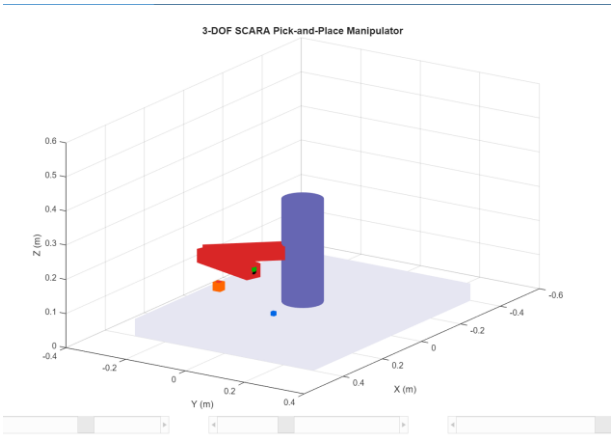
The task sequence follows:

1. Move from home to a safe approach position above the object.
2. Lower vertically to the pick position.
3. Attach the object to the end effector.
4. Lift the object vertically.
5. Move to a safe approach position above the place location.
6. Lower and release the object.
7. Return to the home position.









2. Trajectory Planning

Smooth motion was achieved using high-resolution linear interpolation between task waypoints. This ensured continuous end-effector motion, smooth velocity profiles, no visible jitter or abrupt transitions.

Phase IV: Simulation Implementation

1. MATLAB-Based Simulation

The complete manipulator was simulated in MATLAB with:

- 3D visualization of robot links and joints.
- Integration of forward and inverse kinematics.
- Autonomous pick-and-place control logic.
- Safety checks during motion execution.

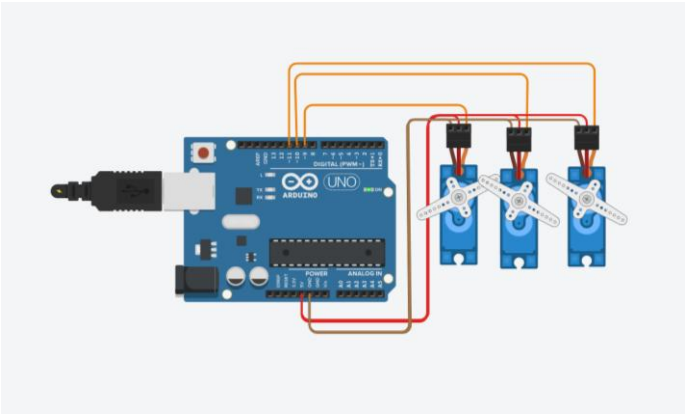
The simulation operates in two distinct modes. Manual joint control using sliders is enabled only after the autonomous task is completed, allowing controlled demonstration without interfering with automation.

2. Arduino and Tinkercad Hardware Simulation

To demonstrate hardware feasibility, the manipulator was also simulated using Tinkercad Circuits with:

- Arduino Uno.
- Three servo motors, each representing one joint of the robot.

Each servo motor is controlled using PWM signals, demonstrating the mapping between kinematic outputs and physical actuators. This validates the feasibility of implementing the simulated system on real hardware.



Results and Observations

- Successful autonomous pick-and-place operation.
- Accurate inverse kinematics performance.
- Smooth and stable robot motion.
- Correct object attachment and release.
- Hardware feasibility validated through Arduino simulation.

Applications

- Industrial pick-and-place systems.
- Assembly automation.

- Educational robotics platforms.
- Manufacturing process simulation.

Conclusion

This project successfully demonstrates the design, kinematic modelling, and autonomous simulation of a 3-DOF SCARA robotic manipulator. The integration of MATLAB-based simulation with Arduino-based hardware modelling provides a clear understanding of robotic manipulation, task planning, and automation principles commonly used in industrial systems.